

Relaxed Randomized Gossip on the Complete Graph: Exact Covariance Spectrum and Sharp Consensus Law

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Abstract

For uniform pairwise relaxed averaging on the complete graph, we derive the exact first moment and sharp mean-square disagreement law at every finite time.

1 Frozen stochastic owner

Fix $N \geq 2$ and $0 \leq \eta \leq 1$. Independently at every integer time, choose an unordered pair $\{i, j\}$ uniformly from the edges of K_N and set

$$x_{t+1} = W_{ij}x_t, \quad W_{ij} = I - \eta d_{ij}d_{ij}^\top, \quad d_{ij} = e_i - e_j. \quad (1)$$

Thus the selected entries become $(1 - \eta)x_i + \eta x_j$ and $\eta x_i + (1 - \eta)x_j$. The endpoint $\eta = 1/2$ is ordinary pairwise averaging, while $\eta = 1$ merely swaps the two entries. The initial vector is deterministic.

Put

$$P = I - \frac{1}{N}\mathbf{1}\mathbf{1}^\top, \quad \bar{x} = \frac{1}{N}\mathbf{1}^\top x_0, \quad y_t = Px_t = x_t - \bar{x}\mathbf{1}.$$

Every W_{ij} is symmetric, fixes $\mathbf{1}$, and commutes with P . The basic complete-graph identity is

$$\sum_{i < j} d_{ij}d_{ij}^\top = NP. \quad (2)$$

Theorem 1 (Mean and sharp energy law). *The sample mean is invariant on every realization. With*

$$\mu = 1 - \frac{2\eta}{N-1}, \quad \rho = 1 - \frac{4\eta(1-\eta)}{N-1}, \quad (3)$$

one has, for every $t \geq 0$,

$$\mathbb{E}y_t = \mu^t y_0, \quad \mathbb{E}\|y_t\|^2 = \rho^t \|y_0\|^2. \quad (4)$$

Proof. Equation (1) fixes $\mathbf{1}^\top x$ pathwise. Since K_N has $N(N-1)/2$ edges, averaging (1) and using (2) gives

$$\mathbb{E}W = I - \frac{2\eta}{N-1}P,$$

which acts by μ on $\mathbf{1}^\perp$. Independence of the new edge from the past proves the first identity by iteration. Since $(d_{ij}d_{ij}^\top)^2 = 2d_{ij}d_{ij}^\top$,

$$W_{ij}^2 = I - 2\eta(1-\eta)d_{ij}d_{ij}^\top.$$

Conditioning on y_t , applying (2), and iterating gives the second identity. It is an equality for every initial vector, hence its rate is sharp whenever $y_0 \neq 0$. $\square$

The phrase *sharp energy owner* records the content added in the original paper round: no spectral conclusion is needed for Theorem 1.

References

- [1] S. Boyd, A. Ghosh, B. Prabhakar, and D. Shah, “Randomized gossip algorithms,” *IEEE Transactions on Information Theory* 52 (2006), 2508–2530. DOI: [10.1109/TIT.2006.874516](https://doi.org/10.1109/TIT.2006.874516).

The source establishes the randomized-gossip owner and network setting. The complete-graph formulas above are derived directly for the frozen model; no literature-priority claim is made for them.